A LINUX DISTRIBUTION FOR TACTICAL INFORMATION AND EMERGENCY COMMUNICATION

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TABLE OF CONTENTS

ABSTRACT ii
ÖZET iii
1. INTRODUCTION 1
2. LITERATURE SURVEY 3
3. AVAILABLE LINUX DISTRIBUTIONS FOR HAMRADIO 4
3.1. Shackbox 4
3.2. FLPuppy 4
3.3. FDGnuX 4
3.4. KB1OIQ - Andy's Ham Radio Linux 4
4. CREATING A LIVE IMAGE 5
5. GENERATING OFFLINE MAP TILES 9
6. FUTURE WORK 10
7. CONCLUSIONS 11
APPENDIX A: LIVE-BUILD CONFIGURATION 12
APPENDIX B: MAKEFILE AND KVM SCRIPTS 13
APPENDIX C: PACKAGE LIST 14
Bibliography 16

ABSTRACT

Amateur radio (also called ham radio) operators play a vital role in providing communication service in emergency situations such as earthquake, fire, and hurricane. Those operators provide different types of communication including voice on different frequencies and data with geographic information using Automatic Packet Reporting System (APRS).

In this work, a live Linux distribution coupled with APRS-aware software that can be used from a USB stick is presented. This distribution focuses on tactical digital communication using APRS and presents a pre-defined configuration of the software before an emergency. Such configuration includes offline map data and set of tools that are useful in an emergency within a USB stick. This USB stick can play a vital role for providing tactical communication during an emergency when there is no Internet connectivity.

ÖZET

Amatör telsizciler (hamradio olarak da adlandırılır), deprem, yangın ve kasırga gibi felaketlerde haberleşme konusunda önemli bir rol oynar. Bu operatörler değişik frekanslarda ses haberleşmesi ile birlikte APRS (Otomatik Paket Raporlama Sistemi) kullanarak coğrafi veri aktarımı da sağlarlar.

Bu çalışmada, içerisinde APRS uyumlu yazılımlar bulunan ve USB bellekten kullanılabilecek bir Linux dağıtımı sunulacaktır. Bu dağıtım APRS kullanılarak taktiksel dijital haberleşme üzerine yoğunlaşmaktadır ve afet sırasında ihtiyaç duyulacak yazılımlar ve önceden indirilmiş haritalar barındırmaktadır. Bu dağıtım afet durumunda internet bağlantısının olmadığı zaman taktiksel haberleşme konusunda önemli bir rol oynayabilir.

1. INTRODUCTION

Amateur radio (also called ham radio) operators play a vital role in providing communication service in emergency situations such as earthquake, fire, and hurricane. In USA, amateur radio operators played a crucial role in providing communication in the following notable events [1]:

- Loma Prieta earthquake, Santa Cruz, California, 17 October 1989
- Oakland/Berkeley fire, October 20-23, 1991
- Hurricane Andrew, Homestead, Florida, 22 August 1992
- Northridge, California earthquake, 17 January 1994

In Turkey, TRAC (Turkish Radio Amateurs Union) provided communication support for governmental and non-governmental organizations in a number of notable events including:

- Marmara Earthquakes, 1999
- Van Earthquake, 2011

Amateur radio operators in those events mostly provided voice communication via Very High Frequency (VHF) and Ultra High Frequency (UHF) channels. However, there is also another communication method called APRS. This system is an amateur radio-based system for real time tactical digital communications of information of immediate value in the local area [2]. Coupled with a Geographic Information System (GIS), APRS can be used to track vehicles, mark locations that are damaged in the event, and send messages using only radios. This system is vital in situations when public phone exchanges, GSM phones, and Internet connection are not usable.

Generally, APRS is used by interfacing a hand-held radio with a computer via USB or serial port. The APRS-aware software reads data from radio and shows the related items in the map. To achieve this, a set of software should be present in the computer (operating system, software, and its configuration). However, it is not possible for a ham radio operator to install this configuration to every laptop present. In an emergency, it is very probable to forget the laptop which has APRS software installed. Nevertheless, it is much easier to find an arbitrary laptop in Disaster Control Centers, which can be used for communication purposes.

In this work, a live Linux distribution coupled with APRS-aware software that can be used from a USB stick is presented. This distribution focuses on tactical digital communication using APRS and presents a pre-defined configuration of the software before an emergency. Such configuration include offline map data and set of tools that are useful in an emergency.

This USB stick is easy to carry while holding enough software to be interfaced with a radio. When disaster happens, it will be enough for ham radio operator to run the USB stick on an arbitrary computer, connect the radio, and start getting tactical information.

2. LITERATURE SURVEY

In *The role of amateur radio in providing emergency electronic communication for disaster management*, Russell C. Coile [1] points out the contribution and importance of hamradio operators in variety of disasters in detail. Those disasters include *Loma Prieta and Northridge earthquakes*, *Oakland/Berkeley fire*, and *Hurricane Andrew*. Additionally, it is pointed out that amateur radio clubs are actively mentioned in *Federal Emergency Management Agency* for disaster management, preparedness, and first response. Moreover, the structure of the crisis management in US is provided.

In *Disaster and crisis management in Turkey: a need for a unified crisis management system*, Unlu, Kapici, and Sahin [3] reviewed the crisis management in Turkey. The general definition of what crisis management involves is given and the reaction of the Turkish government on crisis (mainly earthquakes) is reviewed. After reviewing the law and legislation, it is pointed out that the current crisis management structure is centralized and it slows down the reaction time. In fact, according to them, crisis management should be done in distributed fashion and more control should be given to local authorities. They argue that NGOs (Non-Government Organizations) need to be supported more, including but not limited to Turkish Radio Amateur's union, search & rescue teams, and, Turkish Red Crescent.

3. AVAILABLE LINUX DISTRIBUTIONS FOR HAMRADIO

There are a number of Linux distributions for ham radio operators. However, these distributions do not focus on tactical communications or ease of use but rather they try to bundle all the available software related to ham radio. In this chapter, available Linux distributions will be presented.

3.1. Shackbox

Accordingly to the official website of the project, shackbox is the most complete and downloaded Linux distribution for hamradio operators [4]. It is based on Ubuntu Linux and can be run live from USB stick. It includes many amateur radio applications including SDR, Logging, Rig Control, APRS, SSTV, Echolink, and PSK.

3.2. FLPuppy

FLDigi is a digital modem for Linux that makes use of the sound card [5]. It is widely used by amateur radio operators when interfacing the radio with PC. FLPuppy is a variant of a Puppy Linux, aimed at running FLDigi software on a limited hardware. It is a live Linux distribution that can be used from CD or USB stick.

3.3. FDGnuX

Similar to Shackbox, FDGnuX is a general Linux distribution that bundles a number of amateur radio applications [6]. It seems to be the product of French developers. The main web site of the distribution is in French and their official repository on sourceforge.net is not accessible anymore.

3.4. KB1OIQ - Andy's Ham Radio Linux

Andy's Ham Radio Linux is based on Ubuntu and it includes a number of ham radio software including Fldigi, NBEMS, Gpredict, earthtrack, xcwcp and qrq, XLog and cqrlog, flrig and grig, xnec2c, fl_moxgen, aa-analyzer, owx, VOACAP, glfer, Xastir, gqrx, gEDA, GNU Radio Companion, quisk, direwolf, linamc, FreeDV, wsjt-x, MicroFox 15 Config, and a TinyTrak3 configuration program [7].

4. CREATING A LIVE IMAGE

Debian live build project [8] *(version 4.0.3.-1)* is used to create the live USB disk. The resulting operating system is based on Debian Jessie (8.0). In the first phase, Ubuntu 14.04 was chosen as a distribution but the live build process failed. After the investigation of the build failure, it was realized that live-build package for Ubuntu is maintained by "best effort" basis, meaning that it is not actively maintained but contributors do their best to maintain the package [9]. Inferring from this fact, it became clear that live-build for Ubuntu is not guaranteed to work. For this reason, Jessie was chosen because live build for Jessie is actively maintained by Debian developers and it is used to create live images for Debian project itself. The resulting binary image is written to the usb disk under Ubuntu 14.04 with *dd if=live-build.iso of=/dev/sdX* command where X is the usb device. Configuration file for live-build and the package list can be found in Appendix A and Appendix C respectively.

Live build downloads large amount of packages from debian repositories to create an image. This leads to longer build time and configure-make-build-test cycle. To reduce the amount of time spending on downloading the packages, *apt-cacher-ng* is used to cache packages, which runs at TCP port *3142*. Apt-cacher-ng downloads a package from the remote repository, caches it in local disk, and serves it locally if the package is requested again. Serving already-downloaded package from local disk instead of going to remote repository again significantly improves the build time. To further improve the build time, a debian mirror inside Turkey is specified in apt-cacher-ng. This makes downloading a package faster if it's not found in the cache. Thus, apt-cacher-ng repository address *(http://localhost:3142/ftp.linux.org.tr/debian)* is specified in live-build configuration. Full configuration file for live-build is included in appendix A.

The image comes with Xastir version *2.0.4-2+b1*. Xastir is an APRS client for Linux and it provides *mapping, tracking, messaging, weather, weather alerts, and Search/Rescue features over radio* [10]. It is the main application that is used in the event of an emergency. Xastir can work with a Terminal Node Controller (TNC) to get APRS data from radio and show it on the map. Since its use case is to be run offline, Xastir needs map data to be present to correctly show the items received from the radio. For this reason, an offline map of Turkey in shapefile format is included in the image for use in Xastir. The map is downloaded from geofabrik.de [11], which generates shapefile data from OpenStreetMap (OSM).

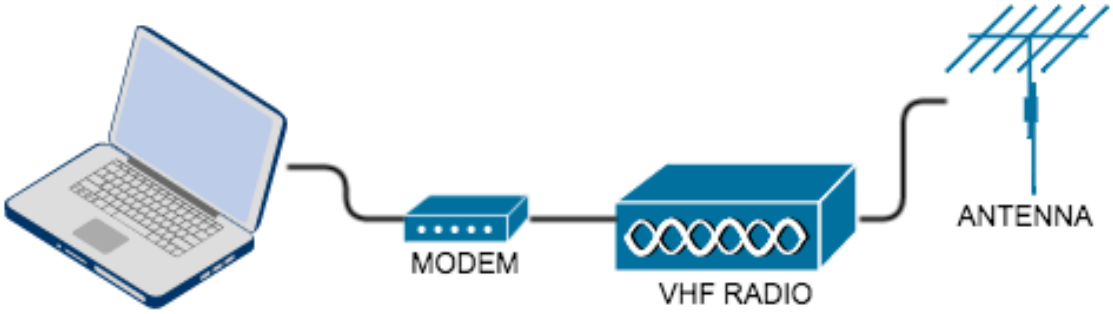

Figure 4.1: An overview of the setup.

APRS data is received using Argent Data OpenTracker+ [12] connected to VHF radio. Keep It Simple, Stupid (KISS) protocol firmware *(version s19)* is used in the tracker. OpenTracker+ acts as a modem between the radio and the computer. Its main function is to demodulate APRS signal received from radio and send digital packets via serial port using KISS [13] protocol. Modern computers, however, do not have a serial port so a serial-to-usb converter is used to interface OpenTracker+ with a computer. After making the required connections and serial port settings, OpenTracker+ is ready for use. With connections to radio and serial port connection to PC using usb-to-serial converter, Xastir is configured to use KISS modem (Opentracker+) as a data source. Xastir initializes the modem, listens for the incoming packets on serial port, renders the items, and shows them in the map. With Xastir set up, an operator is able to get real-time tactical information, mark and send location as well as private messages to other operators. It should be noted that no other connectivity other than radio is present in this configuration.

The default desktop environment (gnome3) is installed in the image for user interface. The current version of the desktop environment is *3.14.0-2*. A shortcut for Xastir which is not present in the official debian package is included on the desktop for easier access to the software.

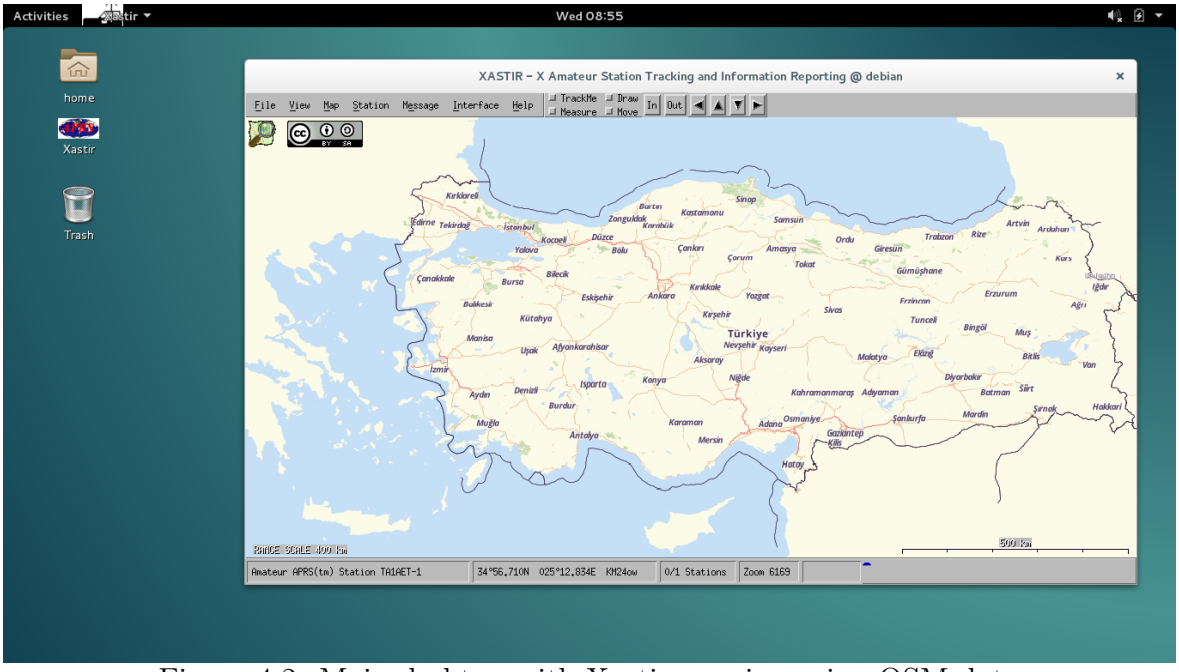

Figure 4.2: Main desktop with Xastir running using OSM data

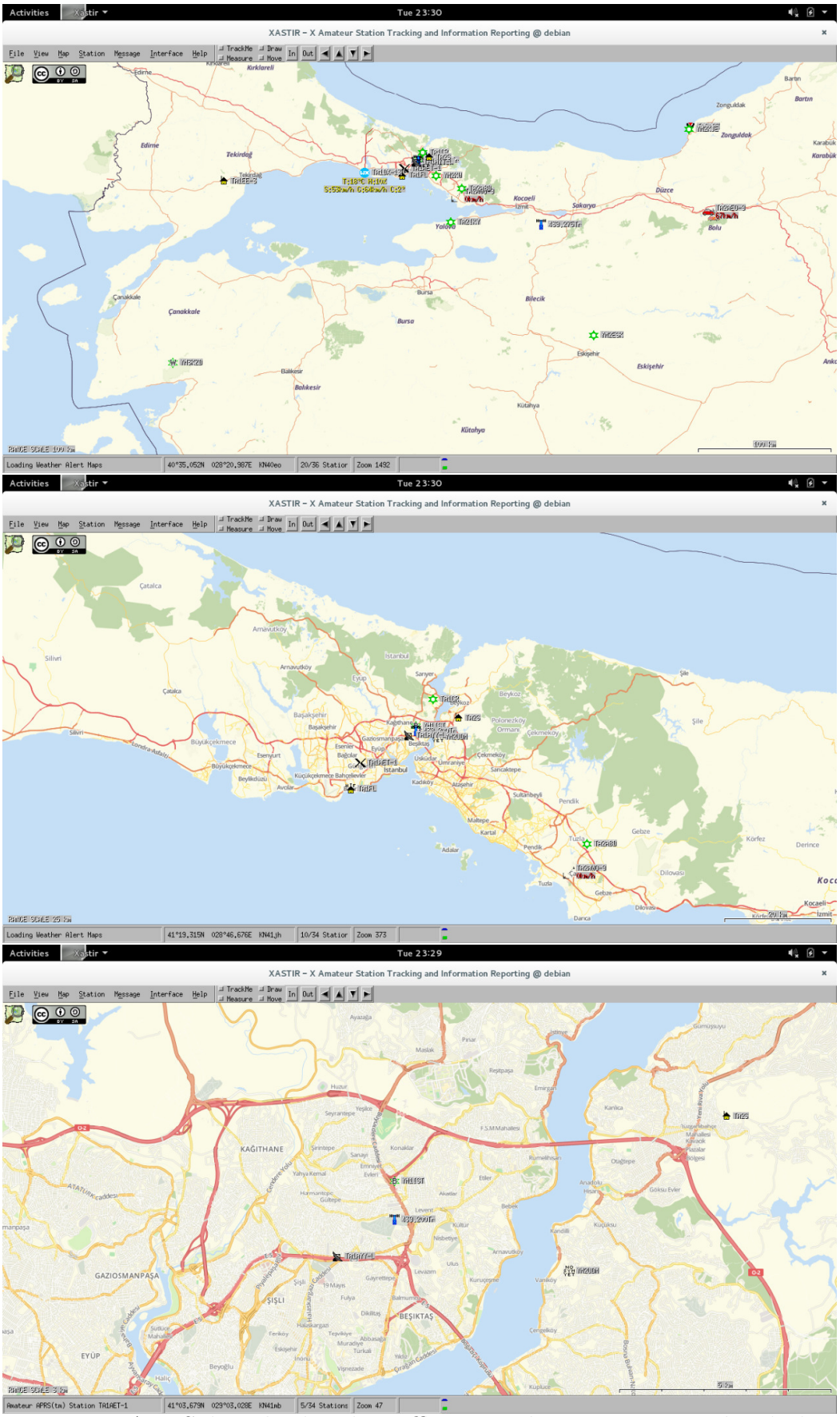

Figure 4.3: APRS data displayed on offline map data using pre-rendered tiles. Different zoom levels are shown.

5. GENERATING OFFLINE MAP TILES

OSM [14] data is used to generate offline maps (tiles). Tiles are 256x256 png files that are displayed by the map software according to coordinates. Those are pre-rendered portions that make up the complete map. Although Xastir can display shapefile data, it has rendering problems which causes Xastir to display roads only as black lines. Hence, pre-rendered images needed to be used in Xastir.

There are many tile servers that can be use download tiles from. However, based on the zoom level and the area, thousands of tiles should be downloaded. For this reason, tile downloaders are usually banned from tile servers. It is, then, required to run a tile server which renders map data and download tiles from local server using a tile downloader.

Tile server is setup with apache2 *(version 2.4.7-1ubuntu4.4)*, mod_tile *(commit rev.774fc7a4470f655)*, and renderd *(commit rev.774fc7a4470f655)* using PostgreSQL database backend *(version 9.3.6)*. Portion of the OSM (shapefile) for Turkey is downloaded from geofabrik.de [11] and processed using *osm2pgsql* utility which parses OSM data and writes to PostgreSQL. After importing OSM data to PostgreSQL, renderd is configured with OSMBright [15] stylesheet. Renderd is a rendering daemon for OSM which interacts with apache/mod_tile and renders the requested map portion using stylesheet. The tile server runs on HP ProLiant G6 Blade System with 24 cores, 24GB CPU and 138GB 15000 RPM disk.

JTileDownloader [16] is used to download tiles from the local tile server. Turkey, as a whole country, is extracted up to zoom level 0-14. Later, west part of Turkey is extracted with zoom level 15. Lastly, Istanbul is extracted with zoom level 16, which gives street-level detail on a map. West part with zoom level 15 had 818,520 tiles alone and it took 2.5 hours to generate/download from the server. It was not feasible to generate tiles with zoom level 16 for the whole country as there are 2 million tiles to be generated. All the tile data is approximately 3.3GB and is copied to the persistent volume on the USB stick for use in Xastir.

6. FUTURE WORK

Persistent volume is planned to be configurable after the first boot. To achieve persistent volume, a user interaction and a working USB disk is required. A user may install the distribution using varying size of USB disks so persistent volume should be handled for different sizes. Hence, a small persistent volume creator tool should be included which operates on the free portion of the disk. An example of such a tool is included in Tails [17], which is a security-focused Debian-based live Linux distribution aimed at preserving privacy and anonymity. It should be noted that persistent volume can be created by manually partitioning the usb disk, formatting the partition with ext3, and creating *persistence.conf* on the partition. However, this method presents a difficulty for the user. It is preferred to have a Graphical User Interface (GUI) that creates persistent volume after the first boot.

The distribution is tested against a small number of configurations. There are different TNCs for APRS and different desktop, laptop configurations. The distribution needs to be tested for variety of configurations. Unfortunately, it was not possible due the the lack of hardware during the study. It should be noted that the distribution is expected to work on every configuration on which Debian Jessie can successfully work.

7. CONCLUSIONS

The available Linux distributions aim to be generic and complete toolkit for amateur radio operators. There is little or no work on tactical information, GIS, or APRS specifically. Although APRS software is included along with other ham radio applications, the main aim of many hamradio distributions is not tactical information but rather being a general toolkit.

This work is aimed to make tactical communication easier by bundling APRS, GIS, and other tools with usability and emergency communication in mind. Effective communication is not possible without location information. This thesis can have an important role in providing tactical information for effective communication and the way we think about emergency situations.

APPENDIX A: LIVE-BUILD CONFIGURATION

```
lb config noauto \
    --mode debian \
    -a i386 \
    -d jessie \
    --archive-areas "main contrib non-free" \
    --parent-archive-areas "main contrib non-free" \
    --mirror-binary "http://ftp.linux.org.tr/debian" \
    --parent-mirror-binary "http://ftp.linux.org.tr/debian" \
    --mirror-chroot "http://localhost:3142/ftp.linux.org.tr/debian" \
    --mirror-bootstrap "http://localhost:3142/ftp.linux.org.tr/debian" \
    --mirror-debian-installer "http://localhost:3142/ftp.linux.org.tr/debian" \
    --parent-mirror-chroot "http://localhost:3142/ftp.linux.org.tr/debian" \
    --parent-mirror-bootstrap "http://localhost:3142/ftp.linux.org.tr/debian" \
    --parent-mirror-debian-installer "http://localhost:3142/ftp.uk.debian.g/debian"
    --security true \
    --updates true \
    --backports true \
    --source false \
    --cache-indices false
```

APPENDIX B: MAKEFILE AND KVM SCRIPTS

```
=> Makefile <==
kvm:
        qemu-system-x86_64 -enable-kvm \
        -name ax25linux -cdrom live-image-i386.hybrid.iso \
        -M pc -m 2048 -cpu kvm64 -vnc :1 \
        -device e1000,netdev=net0,mac=DE:AD:BE:EF:64:6E \
     -netdev tap,id=net0,script=qemu-ifup.sh

isoclean:
        rm -f .build/binary_* .build/chroot_linux-image
        rm -rf binary/isolinux


=> qemu-ifup.sh <==
#!/bin/sh
set -x

switch=br0

if [ -n "$1" ];then
        /usr/bin/sudo /usr/sbin/tunctl -u `whoami` -t $1
        /usr/bin/sudo /sbin/ip link set $1 up
        sleep 1
        /usr/bin/sudo /sbin/brctl addif $switch $1
        exit 0
else
        echo "Error: no interface specified"
        exit 1
fi
```

APPENDIX C: PACKAGE LIST

```
=> cpu.list.chroot <==
### CPU microcodes
amd64-microcode
intel-microcode

==> desktop.list.chroot <==
gnome

ttf-dejavu
fonts-liberation

==> firmware.list.chroot <==
firmware-atheros
firmware-brcm80211
firmware-ipw2x00
firmware-iwlwifi
firmware-libertas
firmware-linux
firmware-linux-free
firmware-linux-nonfree
firmware-ralink
firmware-realtek
b43-fwcutter
firmware-b43-installer
firmware-b43legacy-installer

==> hamradio.list.chroot <==
ax25-tools
ax25-apps
soundmodem
xastir

==> live.list.chroot <==
live-boot
live-config
live-config-systemd

==> peripheral.list.chroot <==
synaptic

==> programming-languages.list.chroot <==
# Programming languages
python
ruby
```

```
==> utils.list.chroot <==
### Serial console
minicom

### Management
ssh

### Downloading and stuff
curl
wget

### Versioning
git
subversion

### Other tools
live-config
unrar-free
pciutils
rfkill
sshfs
resolvconf
pwgen
less
iptables
bash-completion


==> x.list.chroot <==
xserver-xorg-input-all
xserver-xorg-video-all
xserver-xorg-video-qxl

#if ARCHITECTURE i386
xserver-xorg-video-geode
#endif

#if ARCHITECTURE i386 amd64
open-vm-tools
xserver-xorg-input-vmmouse
#endif
```

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